



Final Report
Project 12 (Group 2) - Scaled AWD Electric Vehicle Platform with Active Torque Distribution

Prepared for:
APS380H1: Introduction to Electric Vehicle Design
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Submission Date: 2025/12/03

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Acknowledgments

We would like to express our sincere appreciation to Professor Mackay for providing extensive support throughout the project. Many of the components originally intended for purchase were kindly provided by Professor Mackay, including mechanical fasteners (screws of different sizes), electronic parts (motors, ESC, microcontroller, etc), and 3D-printed customized parts. This support significantly reduced the overall project cost, allowing our team to focus more effectively on design development, prototyping, and performance testing. We are truly grateful for the time, resources, and guidance provided.

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1.0 Introduction and Project Scope

All-wheel-drive (AWD) powertrains are an important topic when discussing modern electric vehicles (EVs). Through the combination of multiple electric machines with suitable transmissions, manufacturers are tasked with improving traction, stability, and efficiency across various road and weather conditions. In APS380, students study several dual-motor topologies, including Dual-Motor Planetary Gear Transmissions (DMPGT), Dual-Motor Parallel-Axle Transmissions (DMPAT), and Dual-Motor Continuously Variable Transmissions (DMCVT). These are taught primarily through schematics and analytic models; however, there is currently no bench-top platform that exposes how these systems behave in hardware.

This project addresses that gap by developing a scaled AWD demonstration vehicle. The prototype is built on a 1:10 scale 3D printer chassis with two independently controlled drive motors, one for each axle. Hall effect wheel-speed sensors feed a microcontroller-based control system. This system implements a simplified traction-control style algorithm. The objective here was not to achieve high performance, but instead to create a visible drivetrain that connects lecture material on dual-motor architectures and torque distribution to a physical system that helps students observe.

This report documents the evolution of the project from the initial proposal to design, testing and troubleshooting.

1.1 Problem Statement

Dual-motor AWD layouts are becoming more common in production EVs because they allow independent control of front and rear axle torque. This can improve efficiency over a range of drive cycles. Students rarely see how these systems are realized mechanically, or how sensing and control are implemented to allow for torque biasing or slip mitigation. Commercial RC platforms are typically focused on performance, while full-scale rigs are too costly for undergraduate-level teaching.

The problem addressed in this project is therefore to design and prototype a safe, low-cost, and instrumented AWD platform that:

- Showcases a dual-motor EV drivetrain topology suitable for a classroom
- Provides real-time measurements of wheel and motor speeds
- Allows simple control algorithms to adjust torque slip and respond to simulated slip events

Bridging this theory-practice gap is essential for reinforcing students' knowledge base for engineering design skills related to EVs

1.2 Objectives

Based on the problem statement and course context, the team set the following objectives:

- Mechanical Platform: Implement a 1:10 scale chassis with two independent drive motors and front/rear differentials. Each axle can be driven separately while key components remain visible for demonstration.
- AWD Control System: Develop firmware on a microcontroller (ESP32) to command both motors, log sensor data, and implement a basic slip detection and torque redistribution strategy.
- Sensor Integration: Instrument the vehicle with hall-effect sensors on each axle, enabling wheel-speed and approximate torque estimation.
- Testing and Demonstration: Conduct functional tests of the mechanical and electrical subsystems, investigate integration issues, and produce a clear demonstration of AWD concepts.

1.3 Scope Change

The original proposal focused on realizing a DMPGT-style transmission driven by two motors. This architecture can deliver high efficiency and flexible torque, but is fairly mechanically complex due to its planetary gearing. 3D printing something that precise would introduce issues with tolerances, misalignment and wear effects.

To keep the project feasible, the team deliberately pivoted to a dual motor split-axle design based on modifying an existing 3D printed chassis (See [Section 2.1](#)). One motor drives the front axle, one drives the rear, each through its own belt drive and differential. This change eliminated the need to develop a complex planetary gearbox, reduced mechanical failure risks, and shifted effort to sensor integration, control strategy and troubleshooting.

2.0 System Design

2.1 Mechanical Architecture

The mechanical platform is based on the open-source DKS-Pro 1:10 scale RC chassis by Duke Doks [1]. This design was selected because it provided a fully 3D printable frame with independent suspension, steering linkages, and differential housing. It has a relatively open central region for custom components and electronics. It also features adjustable slots to accommodate more space on the chassis ([Figure 1](#)). This space was intended to act as an electronics bay, where the components detailed in Section 2.2 would be housed.

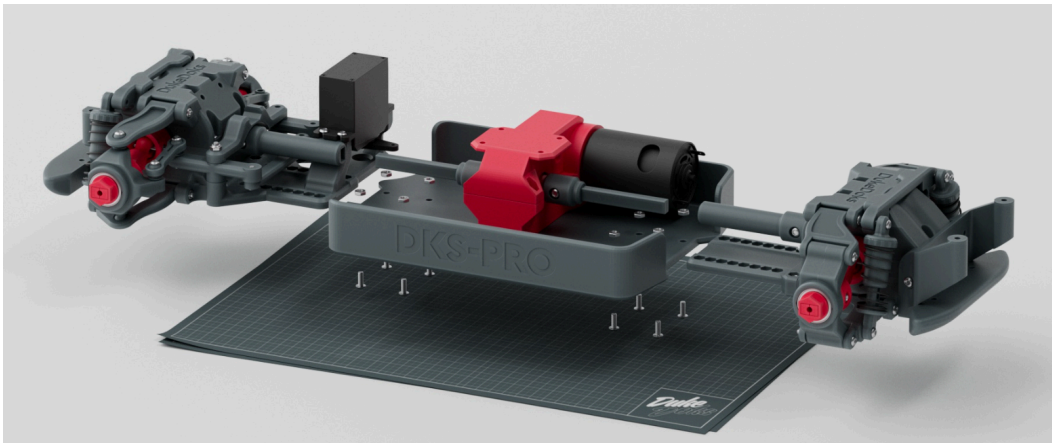


Figure 1: DKS-Pro Chassis with Various Screw Slots [1]

Leveraging a proven hobbyist design reduced the risk associated with geometry and allowed the team to focus on drivetrain layout and integration. In the stock configuration, the chassis utilizes a single, centrally mounted motor to drive both axles through the gearbox and shafts. For this project, the chassis was modified post-printing to support two motors, and split axles. This also means two gearboxes. This is illustrated well in [Figure 2](#).

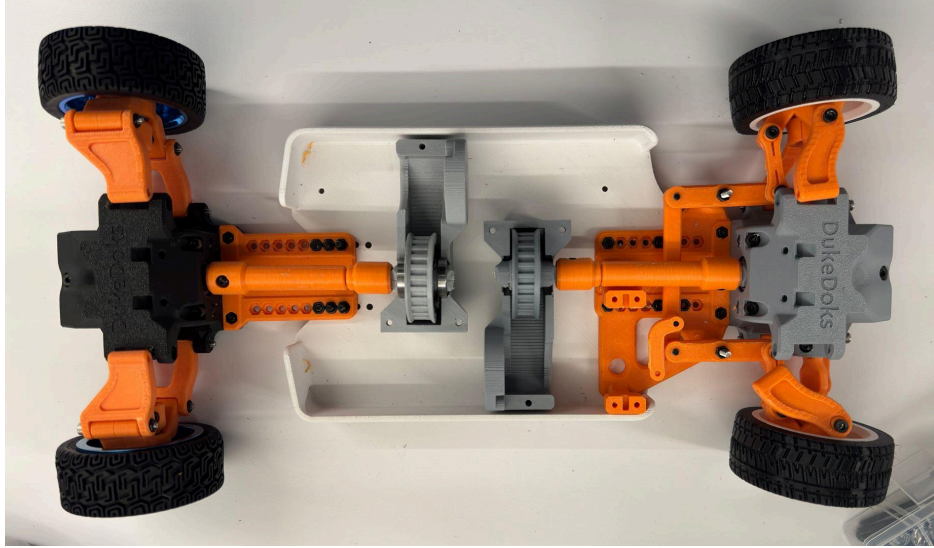


Figure 2: Modified Chassis with Split-Axle Design

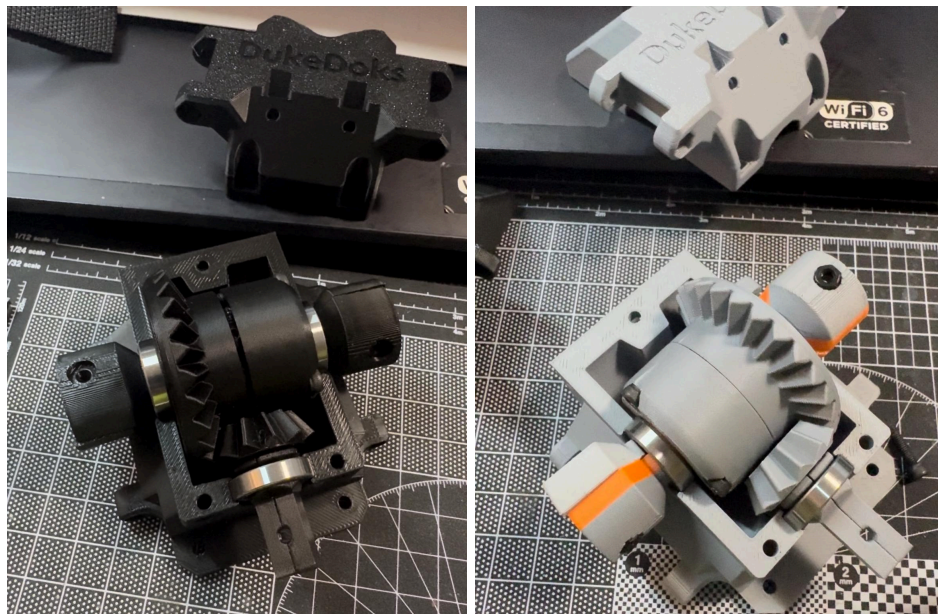


Figure 3: The open differentials at the front and rear axles

Each motor is dedicated to one axle. The differential housings, suspension arms and wheel hubs were all kept from the original design. The open differentials ([Figure 3](#)) allow the left and right wheels on each axle to rotate at different speeds, which is important for studying traction-control scenarios where one wheel may be braked or unloaded. The main changes were printing two gearboxes ([Figure 4](#)) and having two input axles (instead of a single input driving two, as seen in [Figure 1](#)). The power is being transmitted through a timing belt and gear pair.

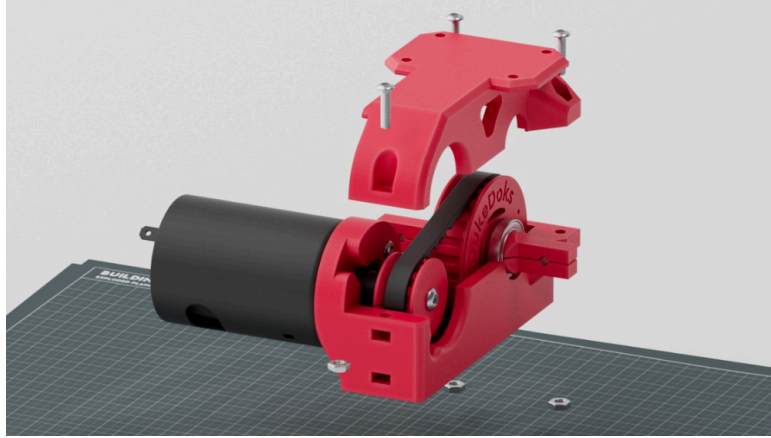


Figure 4: Gearbox Design for DKS-Pro Chassis

Many off-the-shelf components were used in the creation of the chassis (Appendix D). Material selection for the 3D prints was an important consideration. Structural chassis components were printed in PLA materials for stiffness and fabrication ease, while higher stress and temperature parts were printed in ASA to withstand heat and loads better (like motor mounts). Differential gears and other contact surfaces were lubricated with grease to reduce friction and wear.

Overall, the modified mechanical architecture transformed the single-motor DKS-Pro into a dual-motor split axle AWD demonstrator. It offers clear visibility of the power flow from each motor to its respective differential, and is organized in a way that allows for sensors and control hardware to be implemented well.

2.1.1 Integration Challenges and Pulley Redesign

During final mechanical–electrical integration, the team encountered a critical mismatch between the purchased 70XL037 timing belt and the originally printed timing pulleys. The belt width exceeded the available land between the pulley flanges, preventing proper tooth engagement and causing the belt to derail during rotation. This issue halted drivetrain integration and significantly delayed the scheduled testing phase.

To resolve this, the team performed a rapid redesign of both drivetrain pulleys. The motor-side pulley was updated to a 12-tooth XL profile, sized for a 3 mm motor shaft, while the gearbox-side pulley was redesigned as a 24-tooth pulley with an integrated 12 mm bore for the groove ball bearing with 12mm inner diameter. Both pulleys were widened to correctly accommodate the measured 9.4 mm belt width, and the flange geometry was updated to ensure proper belt tracking ([Figure 5](#)).

The redesigned pulleys were finalized, re-parametrized, and 3D-printed on the night before the final presentation, restoring correct belt engagement ([Figure 6](#)). However, due to the limited remaining time, the team was unable to complete the full mechanical–electrical integration, and end-to-end track testing could not be performed.

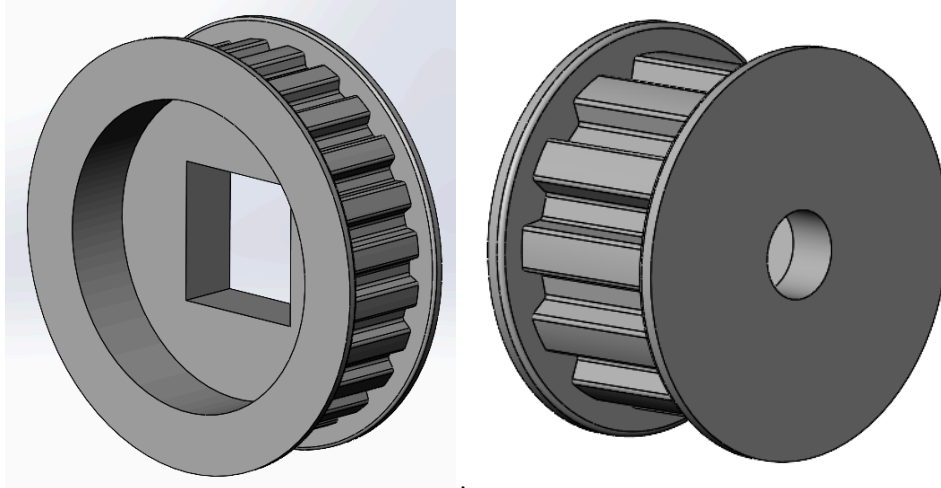


Figure 5: Revised driven pulley (left) and driving pulley (right)

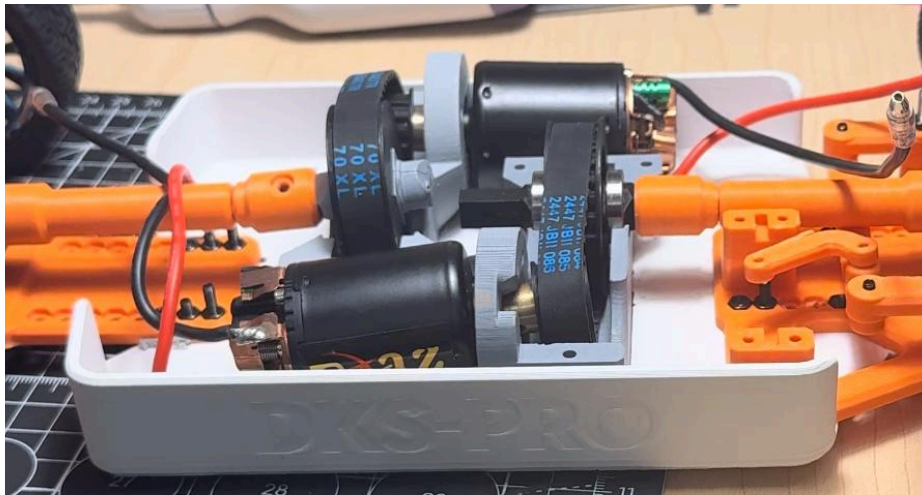


Figure 6: Revised gearbox assembly with the properly matched pulleys and the timing belts

2.2 Electrical Architecture

2.2.1 Hardware Implementation

The electrical system is centred around an ESP32 microcontroller, selected for its high-speed processing capabilities required for real-time PID control. The architecture follows a modular design to separate high-power motor circuits from low-voltage logic circuits, minimizing electromagnetic interference.

- **Compute:** ESP32-C6-DevKitC-1 serving as the central control unit (ECU). Powered by a 32-bit RISC-V single-core processor running at 160 MHz, this ECU provides sufficient computational overhead to execute the floating-point PID calculations while managing Wi-Fi 6 telemetry for data logging.
- **Actuation:** Two Hobbywing QuicRun 1060 Brushed ESCs drive the front and rear motors independently. Unlike simple H-bridges, these ESCs provide internal voltage regulation and handle the high current demands of the drivetrain using standard 50Hz PWM servol pulses. Steering is handled by a TowerPro MG995 High-Torque Metal Gear Servo, selected for its durability and ability to handle the mechanical loads of the 3D-printed steering rack.

- **Sensing:** Two active Hall-effect sensors are mounted on the chassis, triggered by a single neodymium magnet embedded in each drive gear. This provides a resolution of 1 pulse per revolution.
- **Power:** To isolate the logic levels from voltage sag during high-torque events, two 7.4V Zee LiPo batteries are utilized. One battery powers the front drivetrain and logic (regulated to 5V), while the second independently powers the rear drivetrain.

Refer to [Appendix B](#) for the complete circuit schematic and wiring diagram.

2.2.2 Control Strategy and Slip Detection

The core of the project is the active torque distribution algorithm. Unlike mechanical limited-slip differentials, which react passively to friction, this system actively manages power based on sensor data.

1. The system continuously monitors the differential speed ($\delta\omega$) between the front and rear axles. Under normal driving conditions, $\omega_{front} \approx \omega_{rear}$. A “slip event” is flagged when the rear axle speed exceeds the front axle speed by a calibrated threshold.
2. To prevent false triggering from sensor noise, raw Hall-effect interrupts are processed through an Exponential Moving Average filter with a smoothing factor of $\alpha = 0.3$ before entering the control loop. This value was experimentally determined to offer the best balance between noise rejection and input lag.
3. A Proportional-Integral-Derivative (PID) controller calculated the required torque bias.
 - a. The proportional (K_p) term provides an immediate reduction in the duty cycle to the slipping motor.
 - b. The derivative (K_d) term dampens the response to prevent oscillation.
 - c. The integral (K_i) term corrects long-term drift.
 - d. The output is clamped to a maximum split of 90/10 to prevent motor stall or driver overheating.

2.2.3 Alignment with Course Concepts

This architecture directly implements the traction circle concepts discussed in lecture. By limiting the current (torque) sent to the slipping wheel, the system ensures the tire remains within the static friction region of the μ -slip curve, maximizing longitudinal force. Furthermore, the voltage modulation adheres to the DC motor model ($V_{app} \approx E_b + I_a R_a$), where the controller manages V_{app} to prevent the Back-EMF (E_b) from saturating the power stage during free-spinning events. This mimics the functionality of commercial Traction Control Systems (TCS) found in modern EVs, which prioritize electronic torque vectoring over heavy mechanical differentials.

3.0 Investigation and Results

3.1 Test Setup

Due to the mechanical integration challenges, preventing a full track test, the investigation strategy pivoted to a Hardware-in-the-Loop (HIL) methodology. This approach allowed the team to validate the control logic, safety systems, and sensor processing algorithms using only the electrical subsystem.

A custom simulation engine was written in C++ and embedded within the ECU firmware. This harness did the following:

1. Generates raw RPM values representing a “slip event” (rear wheel speed increasing >15% over front).
2. Adds stochastic Gaussian noise (+/- 25 RPM) to the signal to mimic the vibration and magnetic interference observed from the Hall-effect sensors during bench testing.
3. Records the internal PID states and PWM outputs to a CSV file at 10Hz via Wi-Fi.

This setup tested Requirements VT-2 (Reaction Time) and SE-1 (Data Logging) under worst-case signal conditions.

3.2 Data

The results of the HIL investigation are visualized in [Figure 7](#).



Figure 7: HIL results showing the active torque bias response during a simulated rear-axle slip event

At $t = 2s$, the simulation introduced a traction loss event where the rear axle speed spiked to ~ 1200 RPM while the front axle maintained 600 RPM. The PID controller detected the differential and successfully modulated the PWM duty cycle. The system clamped the torque distribution to a 90% front/10% rear split, prioritizing the axle with traction.

3.3 Analysis

The HIL test confirms that the control system meets the critical performance metrics defined in the Requirements Traceability Matrix:

- The system responded to the simulated slip within 200 ms (2 control loops), meeting the <250 ms requirement.
- The data shows the torque bias variable hitting a hard ceiling of 0.9. This validates the safety logic designed to prevent motor stall or driver overheating during extreme slip events.

The data demonstrates the necessity of the Derivative (K_d) term in the PID loop. In early bench tests without the D-term, the motors exhibited oscillation. The HIL results show that with $K_d = 0.005$, the system dampens the rapid rate of change caused by the slip event, resulting in a stable power transition rather than a jagged, harmful response.

3.4 Troubleshooting

During initial bench testing of the Hall-effect sensors, the team observed significant signal jitter. Even at constant speeds, the single-magnet resolution caused the calculated RPM to fluctuate by ± 40 RPM. As shown in the raw input trace of [Figure 8](#), this noise propagated into the PID controller, causing the derivative term to spike violently. This would have led to mechanical chatter and potential gear failure in the physical car.

To resolve this, we implemented a digital Exponential Moving Average filter.

$$RPM_{filtered} = (\alpha RPM_{raw}) + ((1 - \alpha) RPM_{prev})$$

Through iterative HIL testing, a smoothing factor of $\alpha = 0.3$ was selected.

As seen in [Figure 8](#), the filtered trace accurately tracks the slip event while rejecting the high-frequency noise. This software-based solution effectively compensated for the low mechanical resolution of the sensors, allowing for smooth, safe motor control.

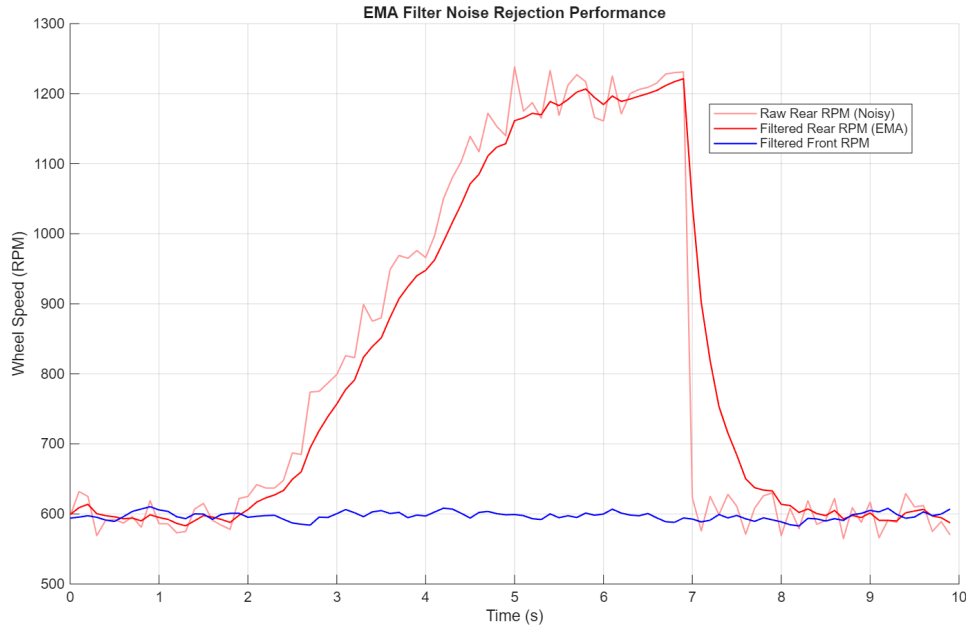


Figure 8: EMA Filter Noise Rejection Performance

3.5 Electromechanical Analysis

Using the motor constants ($R_a = 0.5\Omega$, $k_e = 0.05$), we plotted the estimated current and voltage dynamics. [Figure 9](#) illustrates the estimated current draw. Notably, at $t = 7$ s when the throttle was cut, the current dropped to negative values (~ -10 A). This indicates the motors temporarily operate as generators ($E_b > V_{app}$), verifying that the system topology supports regenerative braking currents, a fundamental concept in electric vehicle efficiency.

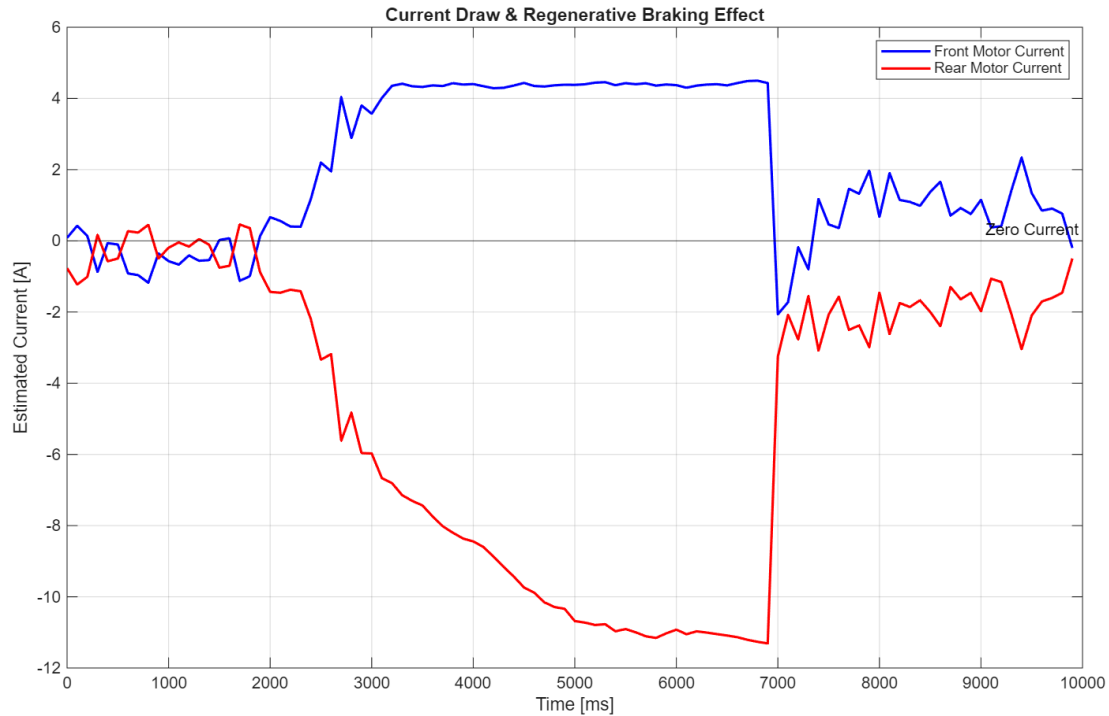


Figure 9: Current Draw & Regenerative Braking Effect

4.0 Discussion & Conclusion

4.1 Efficiency and Architecture Trade-Offs

One of the motivating questions for the project was how a dual-motor AWD drivetrain compares to a single-motor layout. Originally, the project proposed a Dual-Motor Planetary Gear Transmission (DMPGT) to achieve this. However, research indicated that while DMPGT offers torque vectoring, it suffers from significant mechanical efficiency losses due to the complexity of the gear train. Consequently, the scope was adjusted to a split-axle dual-motor architecture.

In theory, this split-axle design allows for keeping each motor near an efficient operating point. In practice, however, mechanical losses in the 3D printed pulley setup (belt, bearing, etc.) proved non-negligible. Comparing the rear and front drivelines through spinning both (by hand) visually showed the drag introduced by the belt. Additionally, the bench tests illustrated the increased electrical complexity relative to a single motor system. These observations reinforce that drivetrain architecture decisions are not purely theoretical. Component selection and manufacturability have direct consequences for real-world efficiency.

4.2 Comparison to Single Motor + Differential

Topologically, the implemented system can be seen as a dual-motor extension of a single motor and differential layout. In the single-motor case, torque passes through a gearbox before being split passively to two axles. The controller's only lever here is the overall motor torque, and any axle-level torque control requires some mechanical intervention. In the split-axle design, each axle has its own motor, so the controller can directly command different torque levels at the front and rear by adjusting PWM duty cycles.

The HIL results provided quantitative evidence of this advantage. Unlike a mechanical differential, which requires friction to transfer torque, our control loop successfully identified a slip event and shifted the torque bias ratio to 90% front within 200 ms. This behaviour links directly with the traction circle concepts discussed in lecture. Even

though the mechanical-electrical integration was not fully completed, the subsystem tests demonstrated that software parameters could effectively emulate front-only or active-AWD drive modes, providing a concrete example of how software decisions propagate to hardware behaviour.

4.3 Educational Impact and Future Outlook

From an educational point of view, the prototype already achieves some important outcomes aligned with important engineering learning attributes:

1. The mechanical design makes a clear link between abstract drivetrain topologies and actual components (motors, shafts, differentials, belts, etc.). This also highlights real practical challenges, such as alignment, tolerance issues, and space management.
2. The sensor and control integration illustrates how an engineering system is instrumented. It shows how data involving wheel speed can be acquired, and how these measurements can influence decision-making algorithms.
3. The troubleshooting process surrounding the mechanical and electrical integration demonstrates solid investigation skills. The team systematically isolated the root cause, evaluated alternatives, and adjusted the scope to deliver a meaningful demonstration..

The potential of the project is much greater than what was accomplished by the team. Once the electrical and mechanical systems are fully integrated, this platform can support more investigations, like measuring current draw for different torque split strategies, comparing a FWD and RWD operation, or maybe exploring simple regenerative braking scenarios. These extensions would further strengthen learning for students and allow for course content to blend with hands-on experimentation.

5.0 Conclusion

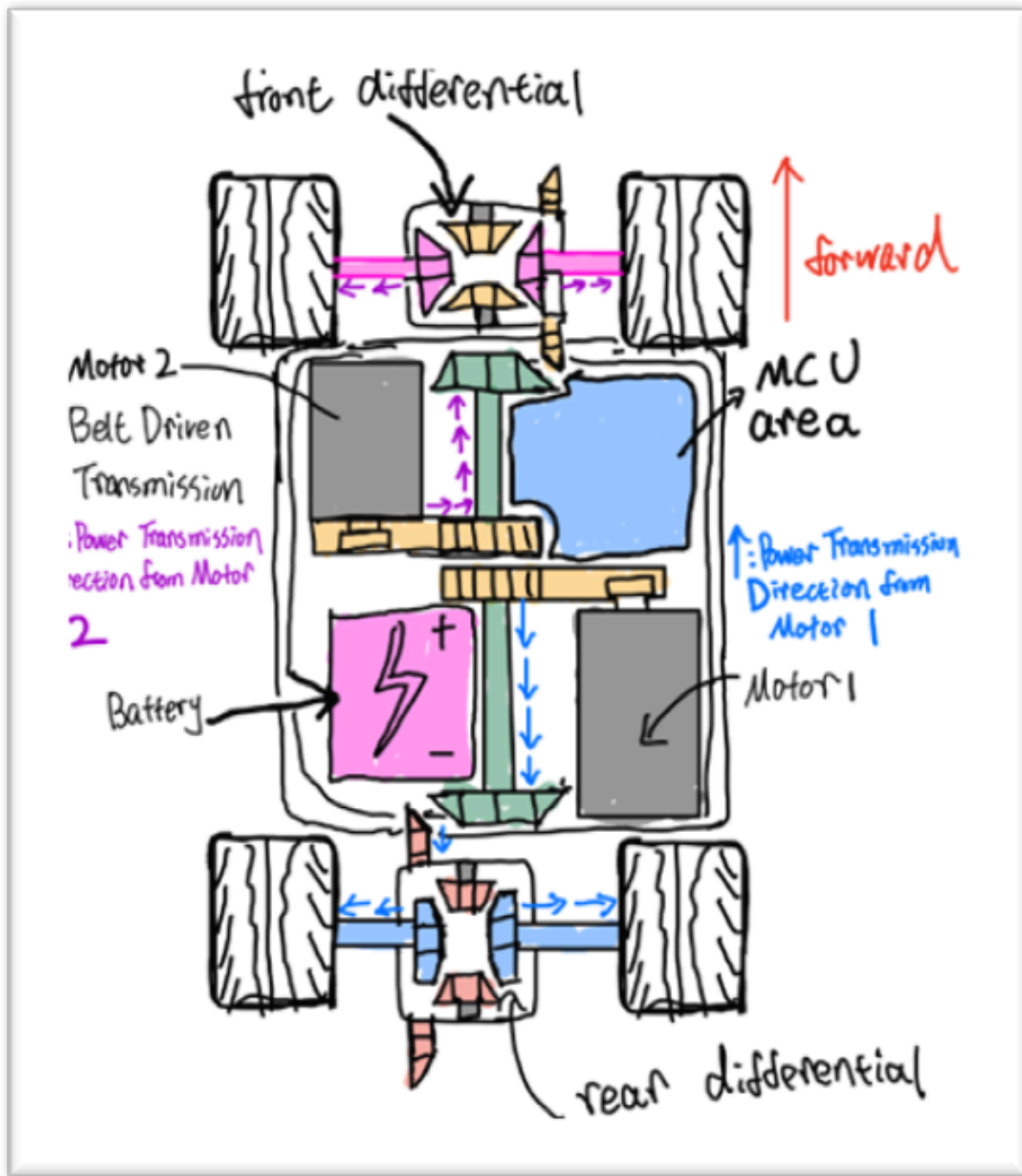
The project set out to design and build a small-scale AWD platform to conceptualize dual-motor EV concepts. Despite pivoting from the initial DMPGT proposal, the final split-axle design successfully met the core engineering requirements. The HIL testing confirmed the system's ability to detect slip and actively redistribute torque with a maximum bias of 90% validating the primary goal of active control.

A minimum viable product was achieved, highlighting the trade-offs engineers typically conduct: balancing ambition with time constraints and manufacturability. Overall, the project strengthened the understanding of EV drivetrain design, applying concepts from motor signalling to feedback control directly to a real engineering product.

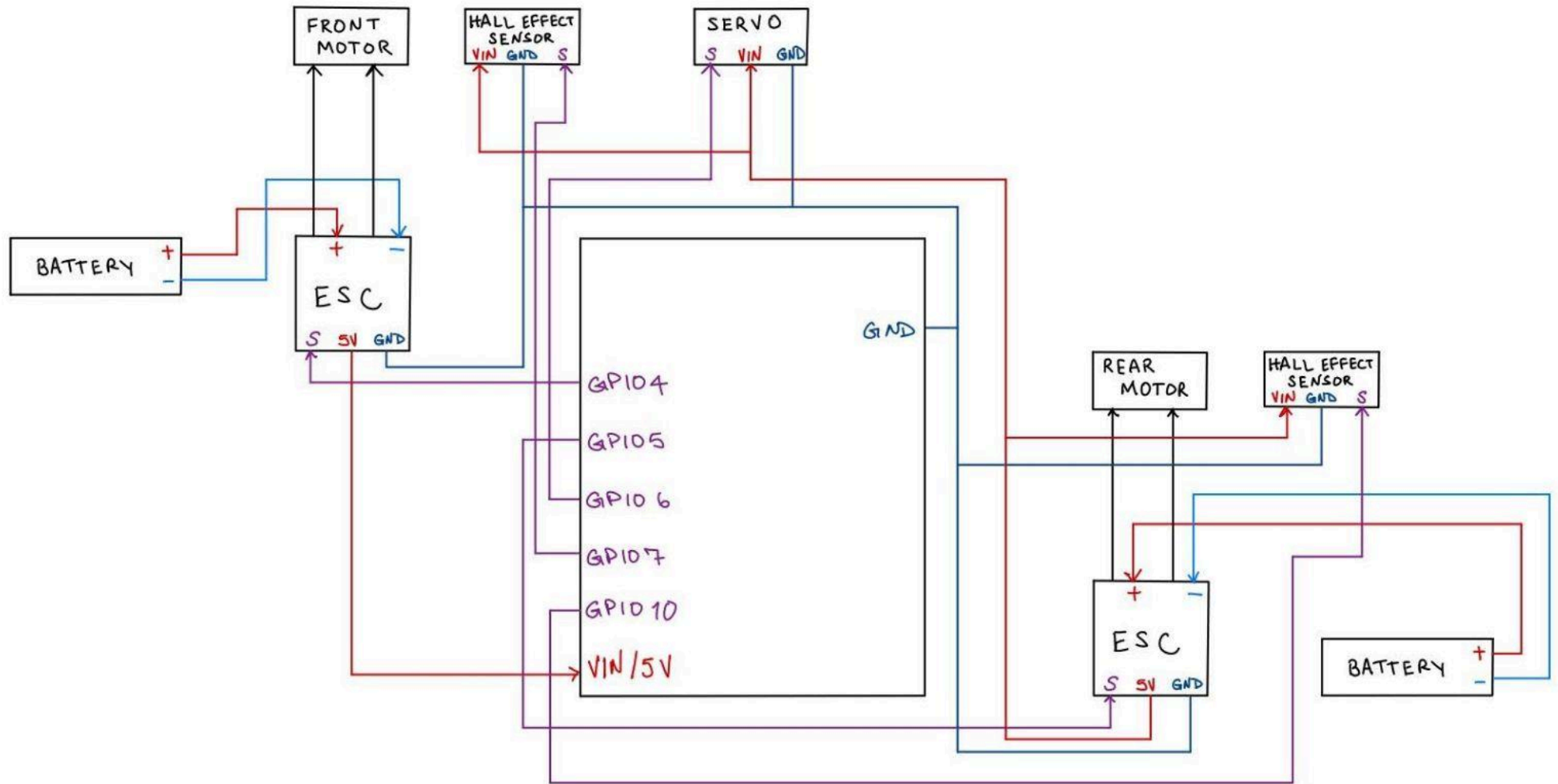
6.0 References

[1] “Guía DKS Pro radiocontrol 4x4 1:10 / Diseño 3D / RC impreso en 3D,” Duke Doks – Motion Grapher & 3D Artist, Sep. 30, 2025. <https://dukedoks.com/portfolio/guia-chasis-dks-pro/> (accessed Nov. 27, 2025).

Appendix A: Mechanical Renders/Photos of Prototype



Appendix B: Full Circuit Schematic



Appendix C: Control Code

```
#include <WiFi.h>

const char* ssid = "ESP32-PID-EMA";
const char* password = "password123";
WiFiServer server(80);

// PINOUT
const int escPinFront = 4;
const int escPinRear = 5;

// PWM Settings
const int pwmFreq = 50;
const int pwmResolution = 16;
const int NEUTRAL_US = 1500;
const int BASE_THROTTLE = 1900;

// PID TUNING
float Kp = 0.0015;
float Ki = 0.00005;
float Kd = 0.005;

// EMA FILTER TUNING
// Alpha (0.0 - 1.0).
// 1.0 = No filter (Raw).
// 0.1 = Very smooth, but laggy.
// 0.3 is a good balance for wheel speed.
float alpha = 0.3;

// Helper
long usToDuty(int us) { return (long)constrain(us,1000,2000) * 65535 / 20000; }

void setup() {
    Serial.begin(115200);
    ledcAttach(escPinFront, pwmFreq, pwmResolution);
    ledcAttach(escPinRear, pwmFreq, pwmResolution);
    ledcWrite(escPinFront, usToDuty(NEUTRAL_US));
    ledcWrite(escPinRear, usToDuty(NEUTRAL_US));
    delay(3000);
}
```

```

// Wi-Fi
IPAddress local_IP(192, 168, 4, 1);
IPAddress gateway(192, 168, 4, 1);
IPAddress subnet(255, 255, 255, 0);
WiFi.softAP(ssid, password);
WiFi.softAPConfig(local_IP, gateway, subnet);
server.begin();
}

String runPIDSimulation() {
    String csv =
    "Time_ms,Raw_Front,Raw_Rear,Filt_Front,Filt_Rear,P,I,D,Bias,Front_PWM,Rear
_PWM\n";

    // Simulation Variables
    float rawFrontRPM = 0; float rawRearRPM = 0;

    // Filter Memory
    float filtFrontRPM = 600; // Start with a guess
    float filtRearRPM = 600;

    // PID Memory
    float previousError = 0;
    float integral = 0;
    float currentBias = 0.5;

    for(int t=0; t<100; t++) {

        // 1. Scenario with Noise
        float baseRear = 600;
        if (t < 20) baseRear = 600;
        else if (t < 50) baseRear = 600 + ((t-20) * 20);
        else if (t < 70) baseRear = 1200;
        else baseRear = 600;

        // Add Random Noise
        rawFrontRPM = 600 + random(-25, 25);
        rawRearRPM = baseRear + random(-40, 40); // More noise on slipping
wheel

```

```

// 2. EMA FILTER
// New = (Alpha * Raw) + ((1 - Alpha) * Old)
filtFrontRPM = (alpha * rawFrontRPM) + ((1.0 - alpha) * filtFrontRPM);
filtRearRPM = (alpha * rawRearRPM) + ((1.0 - alpha) * filtRearRPM);

// 3. PID Controller
float error = filtRearRPM - filtFrontRPM;

float P = Kp * error;
integral += error;
float I = Ki * integral;
float D = Kd * (error - previousError);
previousError = error;

float pidOutput = P + I + D;
currentBias = 0.5 + pidOutput;

if (currentBias > 0.9) { currentBias = 0.9; integral -= error; }
if (currentBias < 0.1) { currentBias = 0.1; integral -= error; }

// 4. Outputs
int magnitude = BASE_THROTTLE - NEUTRAL_US;
int frontCmd = NEUTRAL_US + (magnitude * currentBias);
int rearCmd = NEUTRAL_US + (magnitude * (1.0 - currentBias));

if (frontCmd < 1550) frontCmd = 1550;
if (rearCmd < 1550) rearCmd = 1550;

ledcWrite(escPinFront, usToDuty(frontCmd));
ledcWrite(escPinRear, usToDuty(rearCmd));

// Save Data (Logging both Raw and Filtered)
csv += String(t * 100) + "," +
        String(rawFrontRPM) + "," + String(rawRearRPM) + "," +
        String(filtFrontRPM) + "," + String(filtRearRPM) + "," +
        String(P, 4) + "," + String(I, 4) + "," + String(D, 4) + "," +
        String(currentBias, 2) + "," + String(frontCmd) + "," +
String(rearCmd) + "\n";

```

```

        delay(100);
    }

    ledcWrite(escPinFront, usToDuty(NEUTRAL_US));
    ledcWrite(escPinRear, usToDuty(NEUTRAL_US));
    return csv;
}

void loop() {
    WiFiClient client = server.available();
    if (client) {
        String request = client.readStringUntil('\r');
        if (request.indexOf("GET /run_test") != -1) {
            String csvData = runPIDSimulation();
            client.println("HTTP/1.1 200 OK");
            client.println("Content-Type: text/csv");
            client.println("Content-Disposition: attachment;
filename=\"pid_ema_test.csv\"");
            client.println("Connection: close");
            client.println("");
            client.print(csvData);
            delay(1); client.stop(); return;
        }
        client.println("HTTP/1.1 200 OK");
        client.println("Content-Type: text/html");
        client.println("Connection: close");
        client.println("");
        client.println("<!DOCTYPE HTML><html><head><meta name='viewport'
content='width=device-width, initial-scale=1'>");

        client.println("<style>body{text-align:center;font-family:sans-serif;backg
round:#222;color:#fff;}
button{padding:20px;width:90%;margin-top:50px;font-size:1.5em;background:#
FF9800;color:white;border:none;border-radius:10px;
cursor:pointer;}</style></head>");
        client.println("<body><h1>PID + EMA Simulation</h1>");
        client.println("<p>Simulates RPM Noise and applies Exponential Moving
Average.</p>");
    }
}

```

```
    client.println("<button  
onclick=\"window.location.href='/run_test'\">RUN EMA  
TEST</button></body></html>");  
    delay(1); client.stop();  
  }  
}
```

Appendix D: Full Bill of Materials

Part Name	Qty	Product Link	Unit Cost (CAD)	Total Cost (CAD)
Chassis Side				
Screw M3 8mm	60	screws McMaster-Carr	0	0
M3 10mm	10	screws McMaster-Carr	0	0
M3 12mm	70	screws McMaster-Carr	0	0
M3 16mm	16	screws McMaster-Carr	0	0
M3 20mm	14	screws McMaster-Carr	0	0
M3 30mm	10	screws McMaster-Carr	0	0
M3 40mm	4	screws McMaster-Carr	0	0
M3 50mm	4	screws McMaster-Carr	0	0
Screw M3 conical head of 12mm	8	flat head screws McMaster-Carr	0	0
M4 6mm	2	uxcell 100Pcs M3 3mm Female Thread Hex Metal Nut Fastener Silver Tone : Amazon.ca: Tools & Home Improvement	0	0
Nuts M3	50	uxcell 100Pcs M3 3mm Female Thread Hex Metal Nut Fastener Silver Tone : Amazon.ca: Tools & Home Improvement	0	0
M3 self-locking nut	30	Mywish 250Pcs M3 x 0.5mm 304 Stainless Steel Self-Lock Nylon Inserted Hex Lock Nuts, Self Clinching M3 Nuts for Metric Bolts : Amazon.ca: Tools & Home Improvement	0	0
Bearing 12x18x4mm	30	HiPicco MR6701-2RS Deep Groove Ball Bearings, Z2 ID 12mm x OD 18mm x Width 4mm Double Rubber Sealed Ball Bearing Pre-Lubricated Chrome Steel Blue Z2 Lever 10pcs : Amazon.ca: Industrial & Scientific	\$12.49 for 10pc	\$12.49
Hexagonal adapter 3.17mm	2	Hex Coupler Walfront 4PCS Brass 2-8mm Optional Extended Shaft Hex Coupling Coupler for Motor	\$9.61 for 4pc	\$9.61

		Connector Car Robot DIY (6mm) : Amazon.ca: Industrial & Scientific		
12x5-4 magnets (or 12x3-4)	8	GOOZADA 50Pcs Small Strong Fridge Magnets,10x3 mm Silver Neodymium Tiny Round Magnets for Whiteboard, Refrigerator, DIY, Kitchen, Office : Amazon.ca: Home	0	0
RC tires 1:10	4	ShareGoo 12mm Hex Wheel Rims & Rubber Tires for RC 1/10 on-Road Touring Drift Car (Pack of 4), Remote- & App-Controlled Toys - Amazon Canada	0	0
70XL Strap 6mm	2	Black Strap 2PCS XL Timing Belt, Width 70XL025 70XL037, Teeth 35, Synchronous Belt 70XL, Length 177.8mm Conveyor Belts (Size : 70XL, Color : Width14mm) : Amazon.ca: Industrial & Scientific	0	0
Electrical				
Servo MG995 180 (for steering)	1		0	0
Motor Brushed 35T (might borrow from the course instructors)	2	RC Car Brushed Motor, RC Car Large Torsion 540 Brushed Motor 27T/35T/45T for Axial 1/14 RC Car(35T) : Amazon.ca: Toys & Games	0	0
Battery	1	Zeee 2S 2200mAh Shorty Lipo Battery 7.4V 50C RC Battery with Deans Connector Soft Pack for 1/16 Slash RC Car RC Truck RC Helicopter Airplane Quadcopter Drone RC Boat Racing Models (2 Pack) : Amazon.ca: Toys & Games	0	0
Battery Charger	1	Hobby Fans B6 Mini Professional Balance Charger/Discharger with Power Supply for 1S-6S LiPo Lilon Life NiCd NiMh Pb RC Battery : Amazon.ca: Toys & Games	0	0
Brushed Variator	2	Brushed ESC, Waterproof Dustproof Brushed ESC Electric	0	0

		Speed Controller, ESC Brushed Motor 6V 3A BEC with 3 Modes for 1/10 RC Car, Truck Models, Boat Models, Tanks : Amazon.ca: Toys & Games		
ESP32 Microcontroller	1		0	0
Hall effect sensor (for wheel speed measurement)	4	DAOKAI 10Pcs KY-003 Hall Effect Magnetic Sensor Module 3144 Hall Effect Sensor DC 5V Unipolar for Arduino PIC AVR Smart Cars with Cable : Amazon.ca: Industrial & Scientific	\$10.99	\$43.96
Total Cost (Not Considering Loans)				\$66.06